

KM Pragati Singh

Senior Java Backend Engineer | Spring Boot | Microservices | Kafka | Redis | AWS

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Professional Summary

Java Backend Developer with 4 years of experience building scalable distributed systems using Java, Spring Boot, Microservices, Kafka, Redis, and AWS. Delivered large-scale government and enterprise platforms serving 1M+ users with expertise in event-driven architecture, REST API design, JWT/OAuth2 security, Redis caching, Docker containerization, Drools rules engine, and CI/CD pipelines. Strong advocate of SOLID principles, AI-assisted development, and Agile delivery.

Technical Skills

Languages: Java 11/17, Core Java, SQL

Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Microservices

APIs & Messaging: REST APIs, Apache Kafka, Event-Driven Architecture, OpenAPI 3.0, Swagger, JWT, OAuth2

Rules Engine: Drools (BANMOE), Rule-based Workflow Automation, Procurement Logic Automation

Databases: PostgreSQL, MySQL, JPA/Hibernate ORM, SQL Optimization, Indexing

Caching & Concurrency: Redis (Cache-aside, TTL, Eviction), CompletableFuture, Async Processing

Cloud & DevOps: AWS (S3, EC2), Docker, Kubernetes, Jenkins, CI/CD Pipelines, Git, GitLab, Maven, SonarQube

Security & Testing: Spring Security, JWT, OAuth2, RBAC, JUnit, Mockito, Integration Testing

AI Dev Tools: GitHub Copilot, Cursor AI, Agentic AI

Practices: Agile/Scrum, SOLID, Factory, Strategy, Singleton, Code Review, System Design, JIRA

Work Experience

Tata Consultancy Services (TCS) — Senior Java Backend Developer

July 2022 – Present | Noida, India

Government e-Marketplace (GeM)

Tech Stack: Java, Spring Boot, Kafka, Redis, PostgreSQL, AWS EC2/S3, Docker, Kubernetes, Jenkins, Spring Security, Drools, SonarQube

- Designed and developed Spring Boot microservices for India's largest public procurement platform serving **1M+ government users**; built 30+ standardized REST API contracts using OpenAPI 3.0 with p99 response latency under 300ms across all critical endpoints.
- Implemented Apache Kafka-based event-driven architecture for asynchronous procurement workflows; configured consumer groups, retry handling, and DLQ support processing **50K+ events/day** with fault-tolerant delivery guarantees.
- Integrated Redis cache-aside strategy for high-frequency catalogue and search APIs, reducing PostgreSQL read load by **~40%** and improving average API response time from **~600ms** to **~180ms** under peak traffic.
- Integrated Drools-based Rules Engine (BANMOE) to automate procurement and approval workflows, reducing manual processing effort by **~50%** through rule-based automation across 10+ procurement categories.
- Implemented JWT-based stateless authentication and OAuth2 authorization using Spring Security; enforced RBAC across **20+ secured endpoints** with centralized exception handling and field-level request validation.
- Containerized microservices using Docker and deployed on Kubernetes; automated build, test, and deployment via Jenkins CI/CD pipelines on AWS, achieving **zero critical rollbacks** across production releases.
- Used GitHub Copilot and Cursor AI for code generation, refactoring, and test creation, reducing development effort by **~40%**; applied SOLID principles and design patterns (Factory, Strategy, Singleton) across all service modules.

Tata Play Broadband — User Onboarding & Workflow Platform

Tech Stack: Java, Spring Boot, PostgreSQL, Redis, AWS S3, Jenkins, Spring Security, JWT, SonarQube

- Developed Spring Boot REST APIs for user creation and onboarding modules; implemented Redis caching for session and preference data, reducing repeated DB calls by **~35%** and improving onboarding API throughput significantly.
- Built JWT-secured role-based approval and rejection workflows with Spring Security and RBAC; managed complete request lifecycle (submission → validation → approval → status tracking) across **5+ user roles**.
- Containerized services using Docker and implemented Jenkins CI/CD with SonarQube quality checks, reducing deployment time by **~30%** across release cycles.
- Led backend design discussions, reviewed API contracts, and mentored junior developers; handled production RCA and performance optimizations, achieving zero Sev-1 defects across **10+ consecutive production releases**.

Personal Project

Drools Rule Engine — Loan Approval Automation

Tech Stack: Java 17, Spring Boot, Drools, Redis, PostgreSQL, Docker, JWT, REST APIs

- Built a rule-based loan approval system using Drools evaluating eligibility on credit score, income, and loan type; automated decisions for **~80%** of standard cases, eliminating manual review overhead.
- Implemented Redis caching for rule template lookups, reducing decision processing latency from **~200ms** to **~60ms**; JWT + RBAC secured APIs for admin and applicant roles.
- Containerized with Docker; designed PostgreSQL schemas for application tracking and decision history with composite indexing for sub-100ms status queries.

Education

Bachelor of Technology (B.Tech) — Computer Science & Engineering

2018 – 2022

Dr. A.P.J. Abdul Kalam Technical University, Lucknow